

YOLOv5 Fine-tuning for Object Detection

Dataset, Training
Results, and Inference

윤영진

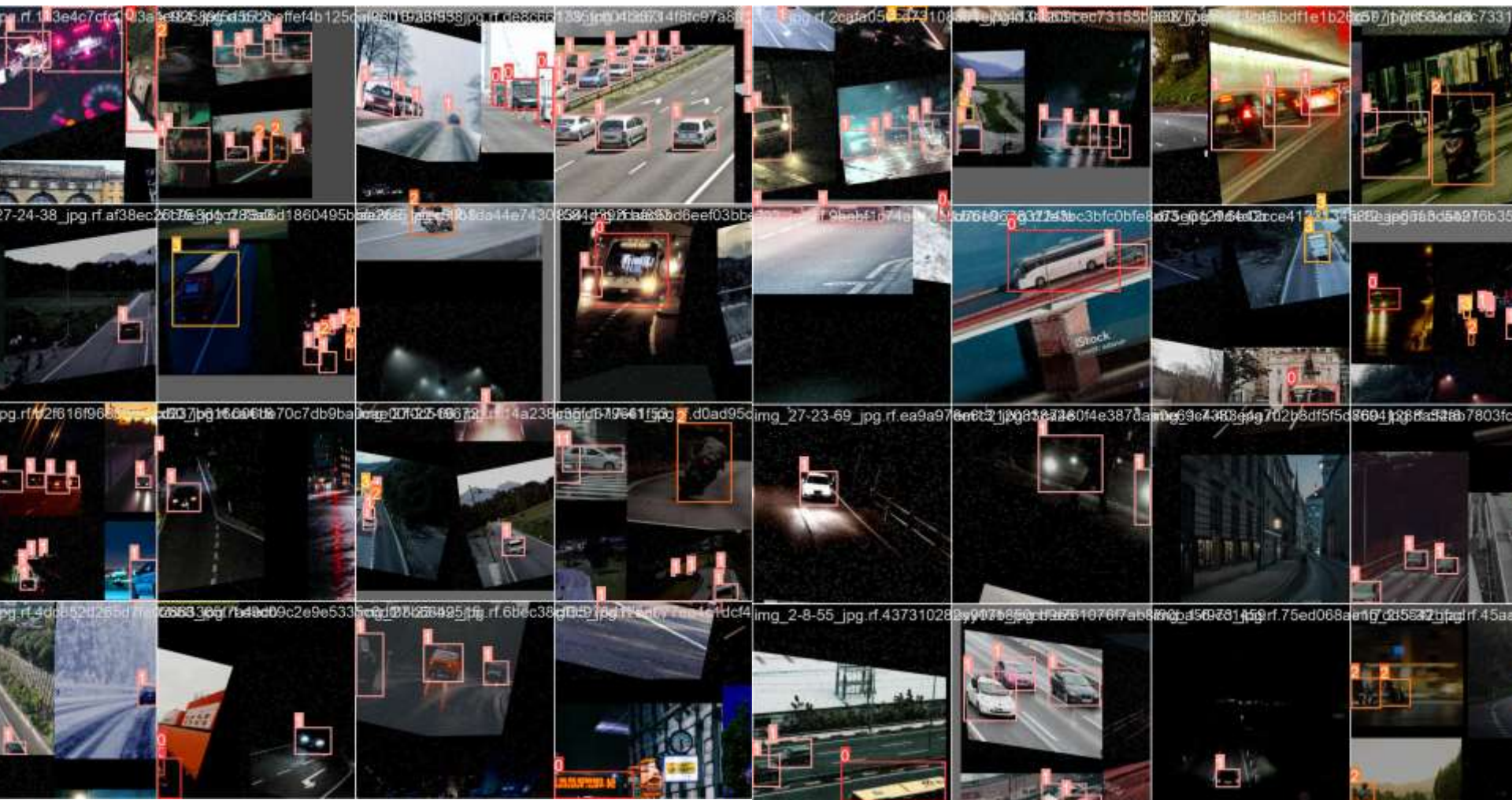
Dataset 소개

- - 출처: Roboflow smart-city-cars-detection(<https://universe.roboflow.com/algorabsmartcity/smart-city-cars-detection/dataset/20>)
- - 클래스 수: 4 ('bus', 'car', 'motorbike', 'truck')

Dataset 소개(train images)



Dataset 소개(train_batch)



Hyperparameters

- - Base model: yolov5m.pt
- - Epochs: 10
- - Batch size: 20
- - Image size: 640

Training 결과

10 epochs completed in 0.493 hours.

Optimizer stripped from src/yolov5/runs/train/detection_yolov5m_xpu/weights/last.pt, 84.0MB

Optimizer stripped from src/yolov5/runs/train/detection_yolov5m_xpu/weights/best.pt, 84.0MB

Validating src/yolov5/runs/train/detection_yolov5m_xpu/weights/best.pt...

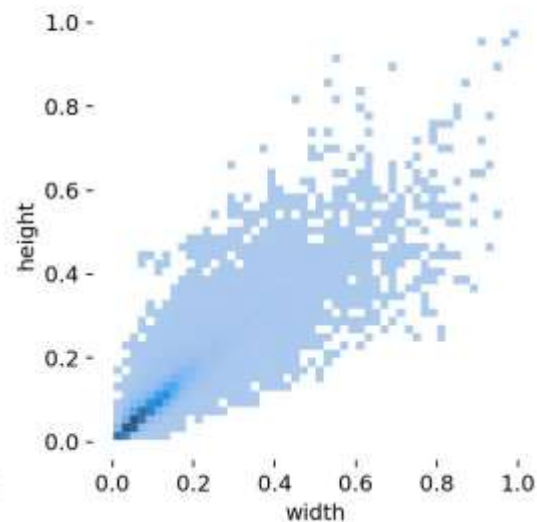
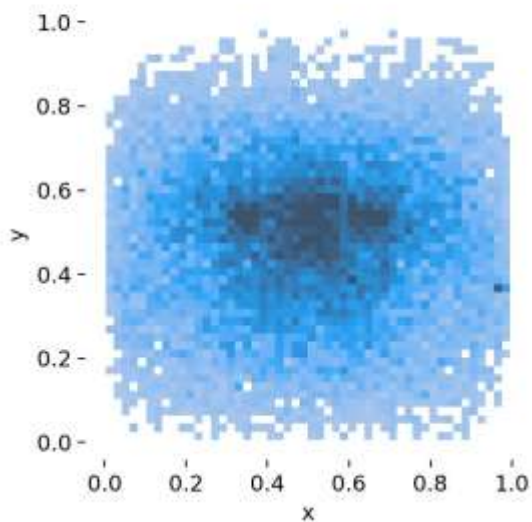
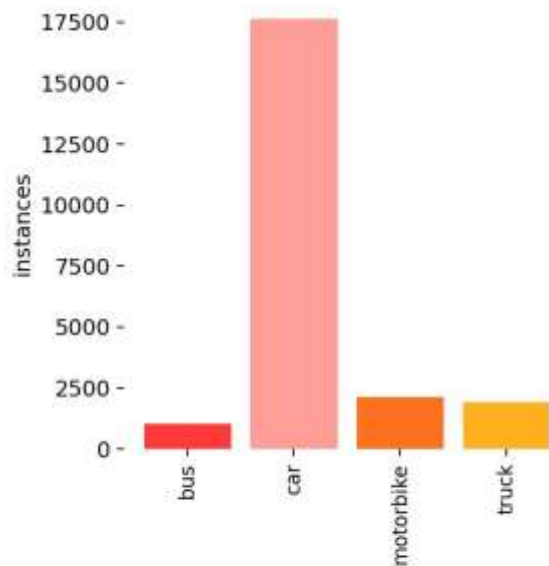
Fusing layers...

Model summary: 212 layers, 20865057 parameters, 0 gradients, 47.9 GFLOPs

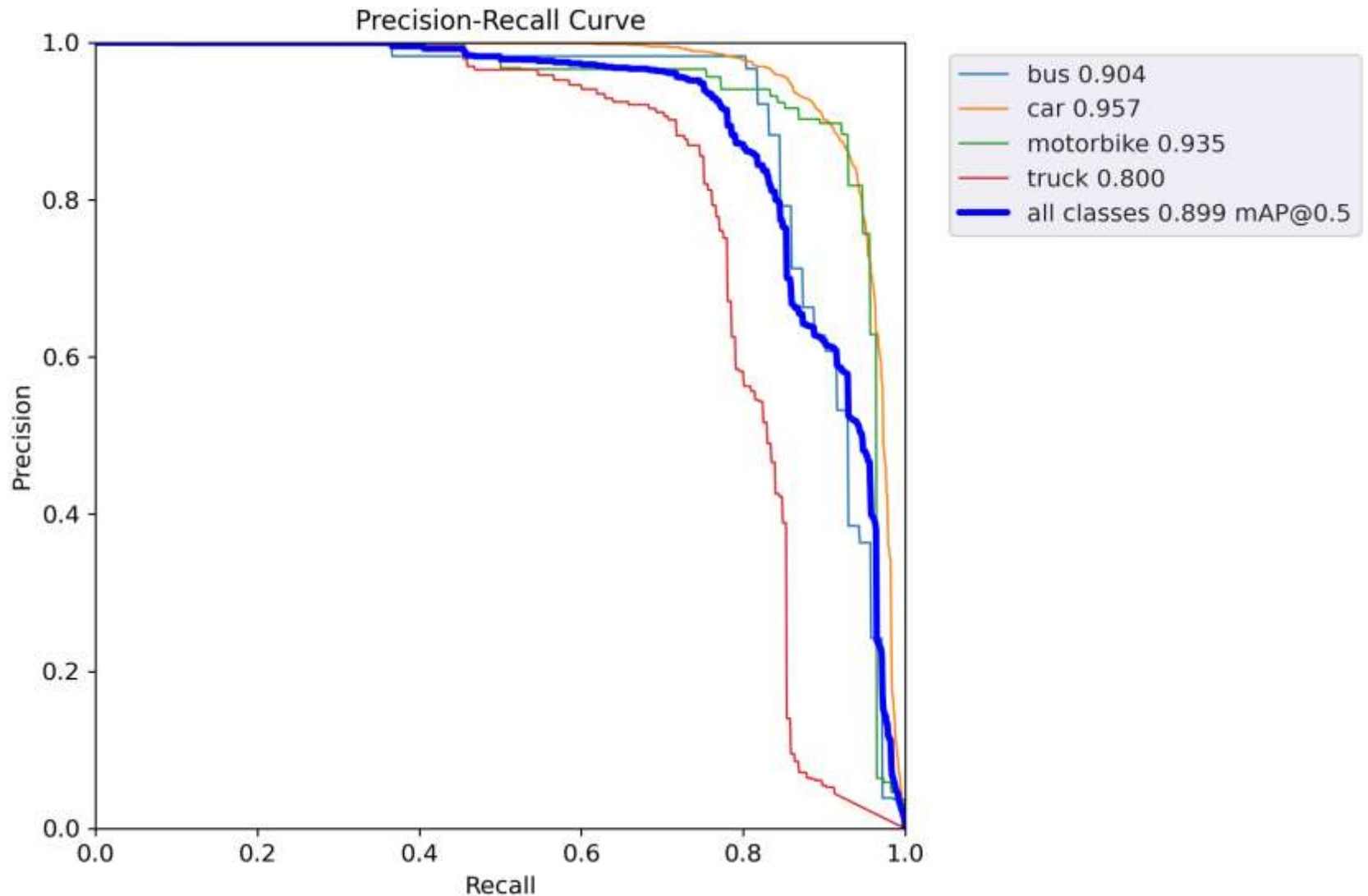
Class	Images	Instances	P	R	mAP50	mAP50-95: 100%
all	432	1860	0.906	0.837	0.899	0.689
bus	432	71	0.976	0.803	0.904	0.735
car	432	1470	0.902	0.899	0.957	0.701
motorbike	432	114	0.895	0.897	0.935	0.713
truck	432	205	0.852	0.751	0.8	0.606

Results saved to src/yolov5/runs/train/detection_yolov5m_xpu

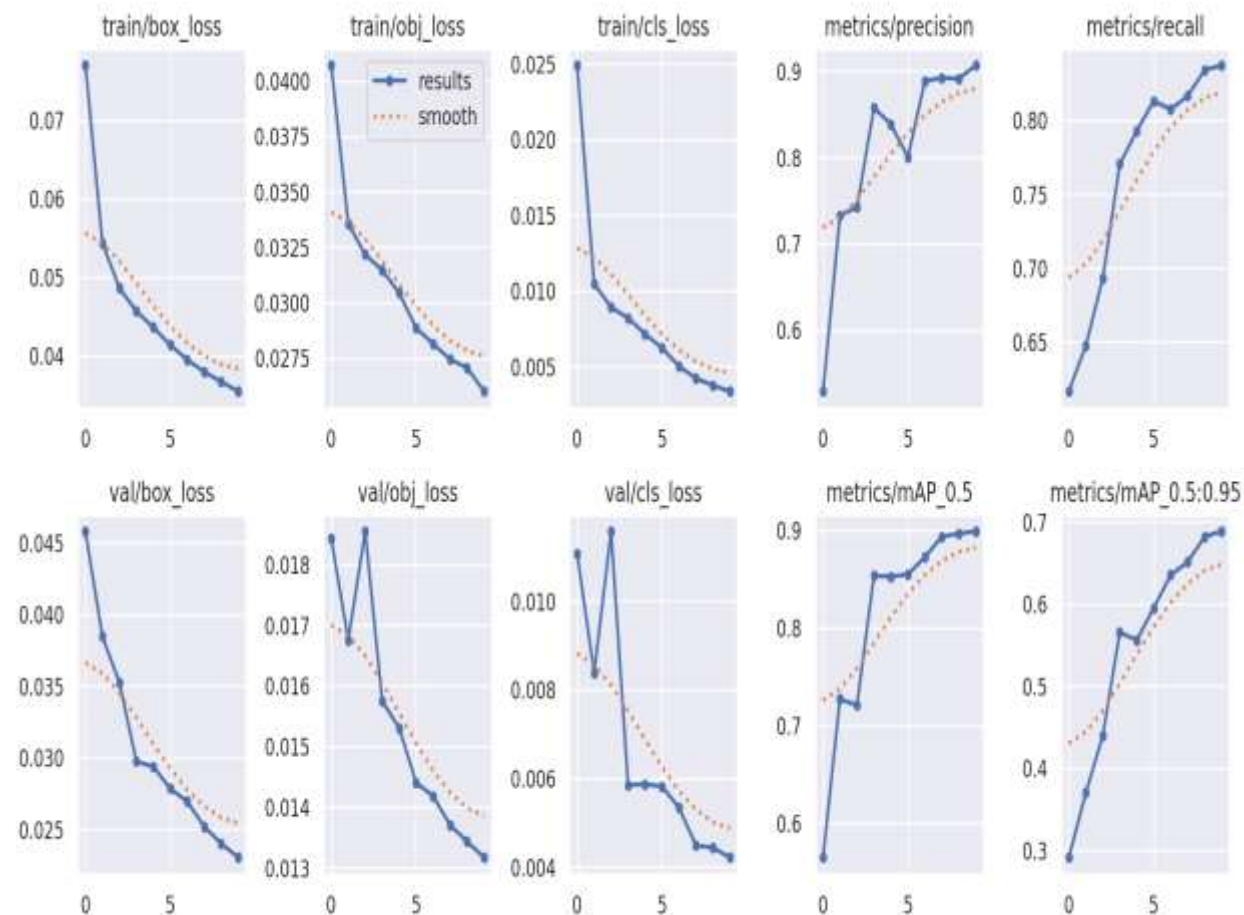
Training 결과



Training 결과



Training 결과



손실값 (train/val box, obj, cls loss)

→ Epoch이 진행될수록 꾸준히 감소 → 학습이 안정적으로 진행됨.

Precision / Recall

→ 초반에 급격히 상승 후 점차 수렴 → 모델이 점점 긍정 샘플을 더 정확히 탐지.

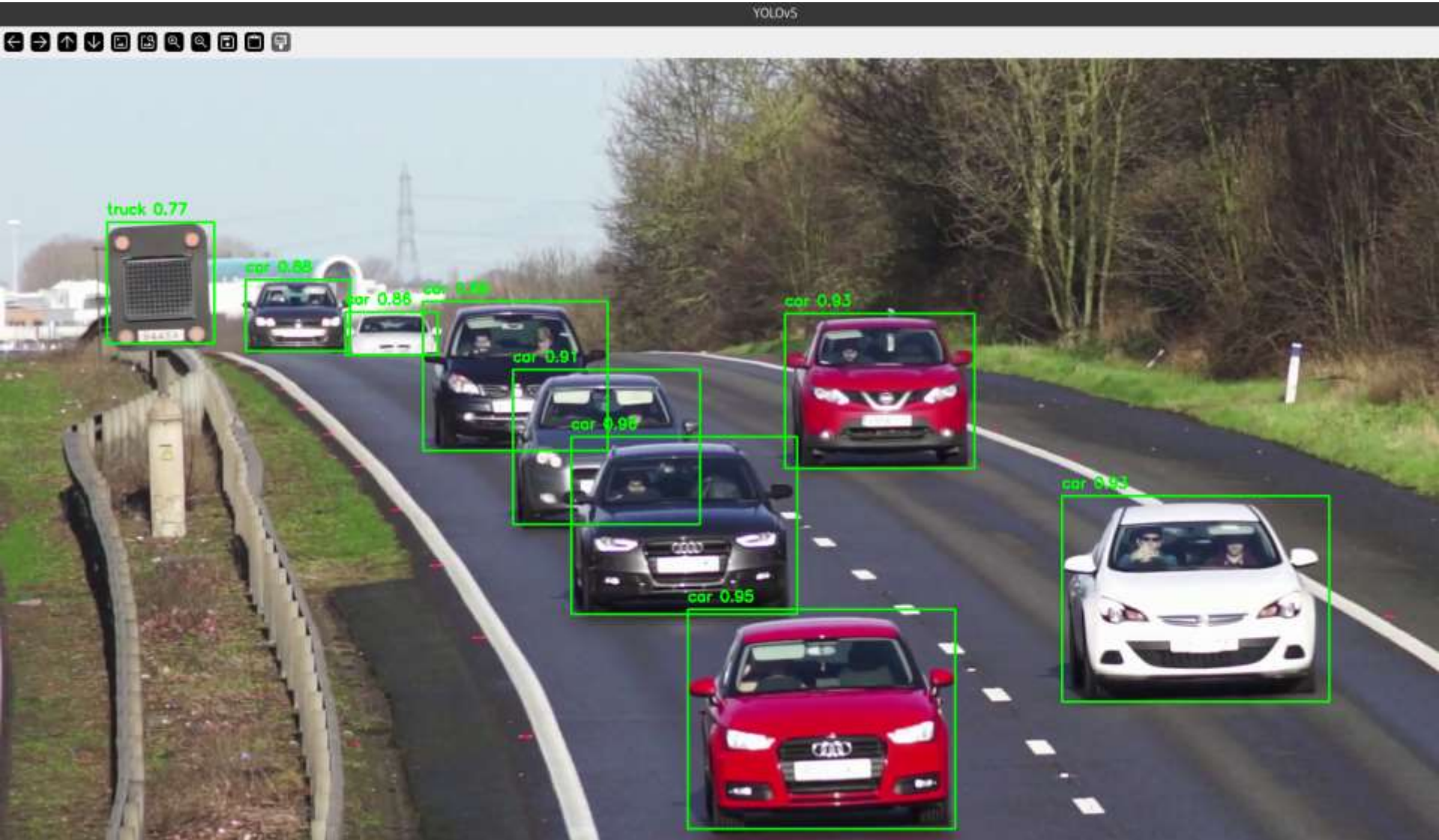
mAP@0.5 및 mAP@0.5:0.95

→ Epoch 증가에 따라 지속적인 향상 → 모델의 전반적인 탐지 성능 개선 확인 가능.

Inference 예제



Inference 예제



Inference 예제



Inference 예제

YOLOv5

